

Proper Categorization

When analyzing algorithmic performance it is important to categorize trades by side (buy/sell/short), size, market conditions (such as up and down days), company capitalization (large, mid, and small cap), etc. Categorization allow analysts to determine if one algorithm works statistically better or worse in certain situations. For example, if VWAP algorithm “A” makes market bets by front-loading executions and VWAP algorithm “B” makes market bets by back-loading, “A” will outperform “B” for buys on days with a positive drift and for sells on days with a negative drift. Conversely, algorithm “B” will outperform “A” for buys on days with a negative drift and for sells on days with a positive drift. A statistical test that combines executions from a large array of market conditions may miss this difference in performance especially if we are comparing averages or medians. It is essential that analysts perform robust statistical hypothesis testing for all performance testing techniques.

Balanced Data Sets

It is imperative that analysts utilize a random selection process for submitting orders to algorithms and ensure that the data sets are balanced across the specified categorization criteria, e.g., size, side, capitalization, market movement, etc. This basically states that the percentage breakdown in the categorization groups described above will be similar. Otherwise, the statistical results may fall victim to Simpson’s Paradox (e.g., dangers that arise from drawing conclusions from aggregate samples).

FINAL NOTE ON POST-TRADE ANALYSIS

One final note on post-trade analysis is the following. Consider the possibility that performance is equivalent across all families of algorithms. For example, there is no difference across VWAP algorithms, IS algorithms, ultra-aggressive algorithms, etc. Subsequently, two important issues arise. First, can brokers still add value to the trading process? Second, is there any need for third party post-trade services? The answer to both these questions is yes.

Brokers can still add value to the process by providing appropriate pre-trade analysis to ensure proper selection of algorithms and algorithmic parameters. Furthermore, brokers can partner with investors to customize algorithms to ensure consistency across the investment and trading decisions. For example, see Kissell and Malamut (2006) and Engle and Ferstenberg (2006). Most